Tasneem Midhat Mustafa Attaallah

Machine Learning Research Engineer

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PROFILE

Machine learning engineer specialising in deep learning, LLM fine-tuning, and scalable retrieval systems. Experienced in building and evaluating embedding-based search pipelines, optimising retrieval accuracy, and developing reproducible ML workflows using Python, PyTorch, FAISS, and FastAPI. Focused on translating AI research into reliable, measurable engineering solutions.

EDUCATION

Lamarr Institute & University of Bonn **Oct. 2026 – Present**
Visiting Student / YAMIR Fellowship

- Focus: Advanced ML Principles, AI Alignment Lab, Advanced topics in LLMs, High-Performance Computing & Parallel Programming.

African Institute for Mathematical Sciences (AIMS) & University of Cape Town Sep. 2025 – Jul. 2026
MSc, AI for Science (Distinction)

- Google DeepMind Scholar (full merit scholarship).
- Coursework: MLOps & Software Engineering, Scientific ML, Bayesian Inference, Tensor Factorisation, CUDA Programming, Explainable AI, Reinforcement Learning, Deep Learning, NLP.

University of Khartoum **2025**
BSc (First Class Honours), Electrical & Electronics Engineering

RESEARCH & ENGINEERING EXPERIENCE

Master's Researcher · *AIMS, co-supervised by InstaDeep & DTU* **2025 – 2026**
INSearch: AI-native retrieval for large-scale proteomic database search `INSearch`

- Architected a scalable dual-encoder retrieval pipeline matching mass spectra to peptide candidates; co-supervised by J. Van Goey (InstaDeep) and Dr. K. Kalogeropoulos (DTU).
- Replaced exhaustive database scans with FAISS/HNSW approximate nearest-neighbour indexing, reducing search complexity to $\mathcal{O}(S(\log P + K))$.
- Trained joint spectrum–peptide embeddings via contrastive learning, VICReg regularisation, and alignment/uniformity objectives, achieving Stage-1 Recall@1/5/100 of 60.0% / 78.9% / 91.9% on a nine-species benchmark.
- Engineered a transformer-based neural re-ranker (InstaNovo decoder), boosting top-1 identification accuracy (Recall@1) to 82.9%.

Software Engineering Intern · *Ajyad Intelligence* **Jan. 2025 – Mar. 2025**

- Developed, tested, and maintained embedded and backend IoT software for industrial smart-sensor deployments.

SELECTED PROJECTS

End-to-End MLOps & Applied AI Applications · `AIMS_course`
Built and deployed modular ML applications following production software engineering standards: structured Python packaging, experiment tracking, automated unit testing, linting/CI workflows, and interactive web/API serving.

`Python · FastAPI/Streamlit · MLflow · Pytest · GitHub Actions (CI/CD) · Docker`

Parameter-Efficient Fine-Tuning (PEFT) for Function Calling & Structured Output
Fine-tuned Llama-3.2-1B-Instruct using LoRA and knowledge distillation to generate schema-compliant JSON and tool/function calls, evaluating robustness across standard leaderboard benchmarks.

`PyTorch · Hugging Face Transformers · PEFT/LoRA · LLM Evaluation`

Recommender System Engine From Scratch · `RecommenderSystemFromScratch`
Engineered a modular recommendation engine in pure Python without high-level RecSys libraries, implementing User-Based and Item-Based Collaborative Filtering (UBCF/IBCF), Matrix Factorisation, and custom vectorised similarity metrics.

`Python · NumPy · Pandas · Linear Algebra · Matrix Factorisation`

High-Performance 5G NR Layer 2 (PDCP) Protocol Stack
Implemented low-latency Packet Data Convergence Protocol (PDCP) packet processing, ROHC header compression, and cryptographic ciphering.

`C++ · C · OpenSSL · Linux`

TECHNICAL SKILLS

Programming: Python, C++, C, SQL, Bash

Deep Learning & AI: PyTorch, Hugging Face (Transformers, PEFT), Contrastive & Representation Learning, LLM Fine-Tuning, Vector Search (FAISS/HNSW), Recommender Systems, Scikit-learn, Explainable AI (SHAP, ALE)

MLOps & Compute: Git, GitHub Actions (CI/CD), Docker, FastAPI, Streamlit/Gradio, MLflow, Pytest, CUDA, Linux

Spoken Languages: English (C1), Arabic (Native)

Eligible for Werkstudent (up to 20h/wk) and mandatory/voluntary Praktikum positions under German regulations (§ 16b AufenthG).